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St. Joseph's college f Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore–607001

B. Sc DEGREE EXAMINATION-APRIL 2021

ABC401-BIOPHYSICS

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions

I. choose the best answer: (10X1=10)

1. The structure of tRNA appears like a

   a) Helix b) Hair pin c) Clover leaf d) Coil

2. Double helical structure model of the DNA was proposed by

   a) Pauling and Corey b) Peter Mitchell

   c) Watson and Crick d) King and Wooten

3. An aromatic amino acid is

   a) Lysine b) Tyrosine c) Taurine d) Arginine

4. The protein present in hair is

   a)Keratin b) Elastin c) Myosin d) Tropocollagen

5. Electrocardiogram is the process of recording the electrical activity of the

   a)kidney b) brain c) heart d) liver

6. EEG is most often used to diagnose the disorders of

   a)liver b) brain c) heart d) kidney

7. Some isotopes decay by emitting positively charged beta particles referred to as

   a) positrons b) negatrons c) protons d) beta rays

8. The sum of protons and neutrons in a given nucleus is called as
   a) atomic number b) nucleus c) mass number d) electrons
9. Emission phenomenon where an energy transition from a higher to a lower state is accompanied by radiation is called
   a) absorption b) fluoroscesnce c) emission d) diffraction
10. Minerals can be estimated by
   a) atomic absorption spectroscopy b) fluorimetry
   c) UV spectroscopy d) X-ray diffraction spectroscopy

II. Say true or false: (5X1=5)

11. The sugar moiety present in DNA is ribose.
12. The number of helices present in a collagen molecule is 3
13. Muscles are composed of two major protein filaments.
14. Unstable isotopes are called radioisotopes.
15. The Beer–Lambert law is valid for higher concentrations only

III. Answer all the questions: (10X1=10)

16. Mention the purine and pyrimidine bases present in the nucleic acids.
17. What is central dogma of molecular biology?
18. Define denaturation of proteins.
19. Write the functions of hemoglobin.
20. What is resting potential?
21. Mention the proteins involved in the muscle contraction.
22. What is decay by negatron emission?
23. Define the radioactive unit curie.
24. Write the principle of UV Spectroscopy.
25. Mention the uses of X-ray diffraction spectroscopy.

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Write a note on different types of RNA.
27. Discuss different types of bonds stabilizing the structure of proteins.
28. Describe the applications of lipoproteins.
29. Write a note on principle and procedure of ECG.
30. Discuss the principle and applications of autoradiography.
31. Write a note on principle and applications of flame photometry.

SECTION-C (3X10=30)

Answer any THREE of the following:

32. Explain the double helical structure of DNA.
33. Describe the biological functions of fibrous and globular proteins.
34. Discuss the molecular basis of muscle contraction.
35. Explain the detection and measurement of radioactivity by scintillation counter.
36. Describe the principle, instrumentation, and applications of spectrofluorimetry.

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